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The Shift from Philanthropy to Measurable Impact

Agricultural CSR in India has undergone a fundamental transformation. Under the current BRSR (Business Responsibility and Sustainability Reporting) framework, Indian corporations are no longer just looking to distribute inputs, they are seeking to build resilient, carbon-neutral ecosystems and enable adoption of regenerative agriculture practices amongst their beneficiary farming communities as a permanent outcome. These outcomes enable organizations to secure carbon credits and enhance their ESG (Environmental, Social, and Governance) ratings using verifiable, science-backed evidence of sustained ecological restoration. However, the transition from chemical-dependent farming to biological alternatives frequently encounters technical barriers, namely, the inconsistent field performance of bio-inputs. These fluctuations can result in diminished yields, often compelling farmers to revert to traditional chemical-intensive practices.

The Operational Reality: Why Traditional Bio-Inputs Fail

Most CSR-led agricultural projects face two recurring issues that deplete their Return on Investment (ROI) and impact metrics:

The Potency Gap: Standard commercial biofertilizers often contain lower than required microbial counts. By the time these products are applied in a remote village, the active "payload" is often too less to compete with native soil pathogens or harsh chemical residues.

The Shelf-Life Crisis: Most biofertilizers come with a 6 to 8-month expiry window. In the context of unpredictable monsoon shifts or delayed distribution schedules common in CSR projects, this leads to high inventory waste and failed field trials.

Our Technical Solution: For a CSR intervention to be successful, the technology must be as robust as the intent. FIB-SOL helps address this issue through a breakthrough in microbial stabilization technology, offering high-density, field-tested biofertilizer and biocontrol formulations designed specifically for the rigors of the Indian rural supply chain.

We have engineered a manufacturing process that enables biofertilizer and biocontrol products containing beneficial soil microbes at an ultra-high density of up to 10¹¹ cells/ml. This ensures that even small dosages deliver a massive biological "boost" to the soil, kickstarting regeneration immediately. Thanks to a proprietary stabilization process, our microbes remain potent for up to 24 months under ambient conditions—eliminating the need for cold-chain logistics and ensuring that your intervention remains viable throughout the project lifecycle.

Aligning Technology with CSR Objectives

Designed to meet the rigorous reporting standards of CSR boards, the outcomes derived from application of our products target essential KPIs including environmental impact, poverty alleviation and community resilience. By aligning our technology with your agricultural CSR framework, we provide a clear pathway toward achieving specific SDGs. A primary advantage of adopting FIB-SOL biofertilizers is the measurable reduction in carbon footprint, providing a verifiable metric for your sustainability disclosures.

Evidence-Based Performance

Our products have undergone rigorous field trials at multiple agri-climatic zones and with multiple crops over the past 5 years. These field trials have clearly established the possibility of enhancing the farming yield by an average of 15% while simultaneously reducing the chemical fertilizer inputs (NPK) by 25%. The plants are found to be healthier, need lesser pesticide interventions and deliver produce with denser nutritional profile. The soil health improves despite the higher yields due to the higher microbial density and their beneficial activity.

Case Study Snapshot

A multi-location, multi-crop trial study of FIB-SOL's Biofertilizer products was carried out in association with Syngenta Foundation for Sustainable Agriculture between Aug 2024 and Jan 2025.

1. Reduction of Nitrogenous(N) Fertilizer (Urea): This study proved that application of FIB-SOL Biofertilizers enables the possibility of enhancing yield even while reducing Urea input as shown in table below.
2. Maintaining Soil Organic Carbon (SOC): This study examined the impact of application of FIB-SOL Biofertilizers on soil health and proved that treatment with FIB-SOL Biofertilizers resulted in significantly increased SOC levels in certain crops namely Tomato, Onion, Cotton, Hot pepper and Eggplant compared to untreated control plots. FIB-SOL treatment enabled maintenance of the SOC levels in crops namely Paddy, Soybean and Maize. This effect of Biofertilizer application improving the SOC levels has been observed worldwide in multiple independent studies. Higher SOC levels in turn results in enhanced water retention in soil and leads to much lower water requirements for agriculture.
3. Improving Farmer Income: This study clearly demonstrates the potential of FIB-SOL products in improving the profitability for farmers due to yield enhancement and reduction in subsidised Urea fertilizer. The study shows that a paddy farmer can earn 12% extra (INR 5300) farm income. A soybean farmer can earn 23% extra (INR 5800) farm income whereas an onion farmer can boost his earnings by up to 14% (INR 16600) depending on the market prices.

Engagement Framework: A Technical Partnership

We recognize that CSR divisions operate through varied models. FIB-SOL has worked with foundations in India such as Syngenta foundation, IFDC, Dr. Reddy's Foundation and few others in the past. As a result, we have acquired robust experience and know-how in implementation of agricultural CSR activities. We have the team structure and partnerships to support you regardless of your implementation style:

1. For Direct Implementers: If your company manages its own rural teams, we act as a technical extension. We provide the high-potency inputs alongside "Train-the-Trainer" modules, ensuring your field officers are equipped to handle soil health diagnostics and application protocols.
2. For Implementation through NGOs: We collaborate with your NGO partners to ensure the scientific integrity of the project. We provide the documentation, soil baseline mapping, and product stability that NGOs need to deliver the results promised in your CSR proposal.
3. Impact Measurement & Reporting: Beyond the product, we provide the data points needed for your annual reports. From quantifying chemical displacement to estimating carbon-saving potential, we help you translate field work into board-room metrics.

Moving Forward

The future of Indian agriculture lies in the soil's biology. To ensure your CSR intervention is both scalable and sustainable, the inputs must be built for the reality of the field.

CONTACT US

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Innovative and Superior Agri-Inputs

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Copy of Biocontrol catalogue

BIOPROTECT Products

Salient Product features-

Ultralight weight - upto 1000X lighter than conventional products

Reduces User Cost: Reduced transport, storage and manpower costs (during application)

High Stability: longer actual shelf life than any product on the market

High Payload of cells: 100X higher than other commercial products

Easy to use: effortlessly works with existing irrigation/fertigation practices

100% eco-friendly: Zero residues and completely biodegradable organic product

PRODUCT CATEGORIES

SUPER SOIL

SUPER SOIL is a highly potent liquid gel formulation that contains >10¹¹ cells/ml of *Pseudomonas fluorescens*

Components: Live Soil Bacteria, Water-soluble biodegradable polymer.

Product Specific USPs

Improves Plant Health and immunity by suppressing soil-borne pathogens

Produces Siderophores and growth hormones to assist Plant growth

Produces Biopolymers such as Chitosan to help maintain soil integrity and texture

Increases germination, nodulation, grain yield and overall productivity.

Mode of Application: Foliar Spraying/Seed coating/Soil Application

SUPER BT

SUPER BT is a highly potent liquid gel formulation that contains > 10¹¹ cells/ml *Bacillus thuringiensis*

Components: Live Soil Bacteria, Water-soluble biodegradable polymer.

Product Specific USPs

Bacillus thuringiensis (or Bt) is a common gram positive, spore-forming, soil bacterium. They produce Cry (Crystal) endotoxins that display insecticidal activity against specific insect groups.

Accumulation of toxin result in pore formation, osmotic cell shock, and ultimately insect death

Insecticidal activity against specific insect groups mainly DBM (Diamond Back Moth) & Helicoverpa. Active against susceptible species in the insect orders Coleoptera, Diptera and Lepidoptera. As well as nematodes (Strongylida, Tylenchida), mites (Acari) and flatworms (Digenea).

Mode of Application: Foliar Spraying

BIO-SHIELD

BIO-SHIELD is a highly potent liquid gel formulation that contains > 1010 spores per ml of Beauveria bassiana

Components: Viable Fungal spores, Water-soluble biodegradable polymer.

Product Specific USPs

It is highly effective natural occurring entomopathogenic fungus that parasites several insects of crops

Beauveria bassiana infects all stages of insect, including eggs, larvae, pupae, grubs, nymphs, hoppers and adult. They cause mortality through a combination of chemical, mechanical, water loss and nutrient loss effects.

Highly active on Lepidopterous Caterpillars including Helicoverpa, Spodoptera, Borers, and Hairy caterpillars. Pest of vegetables and fruits and sucking pest like White fly and Aphids; DBM, Scale insects, Locust, Colorado beetles of Potato.

Mode of Application: Foliar Spraying/Seed coating

SUPER TRI

SUPER TRI is a highly potent liquid gel formulation that contains > 10⁹ cells per ml of Trichoderma viride

Components: Viable Fungal spores, Water-soluble biodegradable polymer.

Product Specific USPs

It is a natural bio-fungicide, highly effective in controlling wide range of soil borne disease caused by Fusarium, Rhizoctonia, Pythium, Sclerotium, Verticillium, Alternaria, Ustilago spp., botrytis cinerea etc.

Growth promotion activity: Suppress soil and seed borne diseases caused by pathogens and promote seed germination and plant growth.

Mode of Application: Foliar Spraying/Seed coating/Soil Application

META-GUARD

META-GUARD is a highly potent liquid gel formulation that contains > 10¹⁰ spores per ml of Metarhizium anisopliae, an entomopathogenic fungus which infects insects, mainly beetle larvae. They penetrate the cuticle and colonizes inside the target pest.

Components: Viable Fungal spores, Water-soluble biodegradable polymer.

Product Specific USPs

It is widely used for control of insects such as root weevils, plant hoppers, Japanese beetle, black vine weevil, spittlebug, termites and white grubs etc.

Mode of Application: Foliar Spraying/Seed coating

VERTI-SHIELD

VERTI-SHIELD is a highly potent liquid gel formulation that contains > 10¹⁰ spores per ml of *Verticillium lilacinus* is an entomopathogenic ascomycete fungus, effective against Aphids.

Components: Viable Fungal spores, Water-soluble biodegradable polymer.

Product Specific USPs

It causes mycosis in the insect orders Homoptera, Coleoptera and Lepidoptera. It produces extracellular enzymes, such as chitinases which helps in promoting host colonization.

Effective against Aphids, mealy bug, Thrips and white flies

Mode of Application: Foliar Spraying/Seed coating

SUPER PAE

SUPER PAE is a highly potent liquid gel formulation that contains > 10⁹ spores per ml of *Paecilomyces*, a cosmopolitan fungus that is mainly known for its nematophagous capacity, but it has also been reported as an insect parasite and biological control agent of several fungi and phytopathogenic bacteria

Components: Viable Fungal spores, Water-soluble biodegradable polymer.

Product Specific USPs

Acts as nematocidal agents, especially against *Meloidogyne* spp., but also against other genera such as *Globodera*, *Rotylenchulus*, *Heterodera*, *Xiphinema* or *Pratylenchus*

Also control fungal wilts caused by *Fusarium* and *Alternaria*

Mode of Application: Foliar Spraying/Seed coating/Root zone spray.